# Continuous Random Variables

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Verifying a valid probability density function**

A continuous random variable $X$ has probability density function $f(x) = \frac{1}{4}$ for $0 \le x \le 4$, and $f(x) = 0$ otherwise. Verify that this is a valid pdf.

Since $f(x) = \frac{1}{4} \ge 0$ throughout, and $\int_0^4 \frac{1}{4}\,dx = \frac{1}{4}(4) = 1$, the total area under $f$ equals $1$, confirming a valid pdf (the uniform distribution on $[0, 4]$).

**2** **[Foundational]** • **Target: Verifying a valid probability density function**

Determine the value of $c$ such that $f(x) = c$ for $2 \le x \le 6$, and $f(x) = 0$ otherwise, is a valid pdf.

$\int_2^6 c\,dx = 4c = 1 \Rightarrow c = \frac{1}{4}$.

**3** **[Foundational]** • **Target: Finding an unknown constant in a pdf**

A continuous random variable $X$ has pdf $f(x) = kx$ for $0 \le x \le 5$, and $f(x) = 0$ otherwise. Find $k$.

$\int_0^5 kx\,dx = k\left[\frac{x^2}{2}\right]_0^5 = \frac{25k}{2} = 1 \Rightarrow k = \frac{2}{25}$.

**4** **[Foundational]** • **Target: Finding an unknown constant in a pdf**

A continuous random variable $X$ has pdf $f(x) = k(4 - x)$ for $0 \le x \le 4$, and $f(x) = 0$ otherwise. Find $k$.

$\int_0^4 k(4-x)\,dx = k\left[4x - \frac{x^2}{2}\right]_0^4 = k(16 - 8) = 8k = 1 \Rightarrow k = \frac{1}{8}$.

**5** **[Foundational]** • **Target: Finding an unknown constant in a pdf**

A continuous random variable $X$ has pdf $f(x) = kx^2$ for $0 \le x \le 3$, and $f(x) = 0$ otherwise. Find $k$.

$\int_0^3 kx^2\,dx = k\left[\frac{x^3}{3}\right]_0^3 = 9k = 1 \Rightarrow k = \frac{1}{9}$.

**6** **[Foundational]** • **Target: Finding probabilities by integrating the pdf**

Using $f(x) = \frac{2}{25}x$ for $0 \le x \le 5$, find $P(X \le 2)$.

$\int_0^2 \frac{2}{25}x\,dx = \frac{2}{25}\left[\frac{x^2}{2}\right]_0^2 = \frac{2}{25}(2) = \frac{4}{25} = 0.16$.

**7** **[Foundational]** • **Target: Finding probabilities by integrating the pdf**

Using the same pdf, find $P(2 \le X \le 4)$.

$\int_2^4 \frac{2}{25}x\,dx = \frac{2}{25}\left[\frac{x^2}{2}\right]_2^4 = \frac{2}{25}(8 - 2) = \frac{12}{25} = 0.48$.

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** **[Intermediate]** • **Target: Finding probabilities by integrating the pdf**

Using $f(x) = \frac{1}{8}(4 - x)$ for $0 \le x \le 4$, find $P(X \ge 2)$.

$\int_2^4 \frac{1}{8}(4 - x)\,dx = \frac{1}{8}\left[4x - \frac{x^2}{2}\right]_2^4 = \frac{1}{8}[(16 - 8) - (8 - 2)] = \frac{1}{8}(2) = 0.25.$

**9** **[Intermediate]** • **Target: Finding probabilities by integrating the pdf**

Using $f(x) = \frac{1}{9}x^2$ for $0 \le x \le 3$, find $P(X \le 1)$.

$\int_0^1 \frac{1}{9}x^2\,dx = \frac{1}{9}\left[\frac{x^3}{3}\right]_0^1 = \frac{1}{9} \cdot \frac{1}{3} = \frac{1}{27} \approx 0.037.$

**10** **[Intermediate]** • **Target: The cumulative distribution function**

For the pdf $f(x) = \frac{2}{25}x$ on $0 \le x \le 5$, find the cumulative distribution function $F(x)$.

$F(x) = \int_0^x \frac{2}{25}t\,dt = \frac{2}{25} \cdot \frac{x^2}{2} = \frac{x^2}{25}$, for $0 \le x \le 5$.

**11** **[Intermediate]** • **Target: The cumulative distribution function**

Using $F(x) = \frac{x^2}{25}$, find $P(X \le 3)$ and hence $P(X > 3)$.

$F(3) = \frac{9}{25} = 0.36$. $P(X > 3) = 1 - 0.36 = 0.64$.

**12** **[Intermediate]** • **Target: The cumulative distribution function**

For the pdf $f(x) = \frac{1}{8}(4 - x)$ on $0 \le x \le 4$, find the cumulative distribution function $F(x)$.

$F(x) = \int_0^x \frac{1}{8}(4 - t)\,dt = \frac{1}{8}\left(4x - \frac{x^2}{2}\right) = \frac{x}{2} - \frac{x^2}{16}$. Checking $F(4) = 2 - 1 = 1$ confirms the total probability is correctly $1$.

**13** **[Intermediate]** • **Target: The cumulative distribution function**

For the pdf $f(x) = \frac{1}{9}x^2$ on $0 \le x \le 3$, find the cumulative distribution function $F(x)$.

$F(x) = \int_0^x \frac{1}{9}t^2\,dt = \frac{1}{9} \cdot \frac{x^3}{3} = \frac{x^3}{27}$.

**14** **[Intermediate]** • **Target: Using the cdf to find probabilities**

Given $F(x) = \frac{x^2}{25}$ for $0 \le x \le 5$, find $P(1 \le X \le 4)$.

$F(4) - F(1) = \frac{16}{25} - \frac{1}{25} = \frac{15}{25} = 0.6$.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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**15** [Challenging] • Target: Using the cdf to find probabilities • Cross-Lesson: 14.01 Sampling & Estimation

Using $F(x) = \frac{x}{2} - \frac{x^2}{16}$, find $P(1 \le X \le 3)$.

$F(3) = \frac{3}{2} - \frac{9}{16} = \frac{15}{16}$. $F(1) = \frac{1}{2} - \frac{1}{16} = \frac{7}{16}$. $P(1 \le X \le 3) = \frac{15}{16} - \frac{7}{16} = \frac{8}{16} = 0.5$.

**16** [Challenging] • Target: Using the cdf to find probabilities • Cross-Lesson: 15.01 Hypothesis Testing

Using $F(x) = \frac{x^3}{27}$ for $0 \le x \le 3$, find $P(1 \le X \le 2)$.

$F(2) - F(1) = \frac{8}{27} - \frac{1}{27} = \frac{7}{27} \approx 0.259$.

**17** [Challenging] • Target: Expectation of a continuous random variable • Cross-Lesson: 2.01 Exponentials & Logarithms

Using $f(x) = \frac{2}{25}x$ for $0 \le x \le 5$, find $E(X)$, using $E(X) = \int x f(x)\,dx$.

$E(X) = \int_0^5 x \cdot \frac{2}{25}x\,dx = \frac{2}{25}\left[\frac{x^3}{3}\right]_0^5 = \frac{2}{25} \cdot \frac{125}{3} = \frac{10}{3} \approx 3.333$.

**18** [Challenging] • Target: Expectation of a continuous random variable • Cross-Lesson: 3.01 Advanced Trigonometry

Using the uniform pdf $f(x) = \frac{1}{4}$ for $0 \le x \le 4$, find $E(X)$, and verify it matches the midpoint of the interval.

$E(X) = \int_0^4 x \cdot \frac{1}{4}\,dx = \frac{1}{4}\left[\frac{x^2}{2}\right]_0^4 = \frac{1}{4}(8) = 2$, matching the midpoint of $[0, 4]$, as expected for a uniform distribution.

**19** [Challenging] • Target: Expectation of a continuous random variable • Cross-Lesson: 4.01 Differentiation

Using $f(x) = \frac{1}{9}x^2$ on $[0, 3]$, find $E(X)$.

$E(X) = \int_0^3 x \cdot \frac{1}{9}x^2\,dx = \frac{1}{9}\left[\frac{x^4}{4}\right]_0^3 = \frac{1}{9} \cdot \frac{81}{4} = \frac{9}{4} = 2.25$.

**20** [Challenging] • Target: Variance of a continuous random variable • Cross-Lesson: 5.01 Integration

Using $f(x) = \frac{2}{25}x$ on $[0, 5]$ and $E(X) = \frac{10}{3}$ (found earlier), find $\mathrm{Var}(X)$, using $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$.

$E(X^2) = \int_0^5 x^2 \cdot \frac{2}{25}x\,dx = \frac{2}{25}\left[\frac{x^4}{4}\right]_0^5 = \frac{2}{25} \cdot \frac{625}{4} = 12.5$. $\mathrm{Var}(X) = 12.5 - \left(\frac{10}{3}\right)^2 = 12.5 - \frac{100}{9} \approx 1.389$.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Probability Density Functions (PDF)

A continuous random variable $X$ has PDF $f(x) = kx(2 - x)$ for $0 \le x \le 2$, and 0 otherwise. Find the value of $k$.

**a.** $\frac{3}{4}$ ✓

**b.** $\frac{1}{2}$

**c.** $\frac{3}{8}$

**d.** 1

Total probability must equal 1: $\int_0^2 k(2x - x^2)dx = k[x^2 - x^3/3]_0^2 = k(4 - 8/3) = k(4/3) = 1 \Rightarrow k = 3/4$.

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**22** [Foundational] • Target: Probability Density Functions (PDF)

For the random variable in the previous question ($k = 3/4$), calculate $P(0 \le X \le 1)$.

**a.** $\frac{1}{2}$ ✓

**b.** $\frac{3}{8}$

**c.** $\frac{1}{4}$

**d.** $\frac{5}{8}$

$P(0 \le X \le 1) = \frac{3}{4}\int_0^1 (2x - x^2)dx = \frac{3}{4}[x^2 - x^3/3]_0^1 = \frac{3}{4}(1 - 1/3) = \frac{3}{4}(2/3) = \frac{1}{2}$.

**23** [Foundational] • Target: Probability Density Functions (PDF)

What is the probability that a continuous random variable takes any single exact value, such as $P(X = c)$?

**a.** $0$ ✓

**b.** $1$

**c.** $f(c)$

**d.** $f'(c)$

For any continuous distribution, the probability of taking any exact single point is zero ($P(X = c) = \int_c^c f(x)dx = 0$).

**24** [Foundational] • Target: Cumulative Distribution Function $F(x)$

Given PDF $f(x) = \frac{3}{8}x^2$ for $0 \le x \le 2$, find the cumulative distribution function $F(x)$ for $0 \le x \le 2$.

**a.** $F(x) = \frac{x^3}{8}$ ✓

**b.** $F(x) = \frac{3x^3}{8}$

**c.** $F(x) = \frac{x^2}{4}$

**d.** $F(x) = \frac{x^3}{2}$

$F(x) = \int_0^x f(t)dt = \int_0^x \frac{3}{8}t^2 dt = \left[\frac{t^3}{8}\right]_0^x = \frac{x^3}{8}$.

**25** [Foundational] • Target: Cumulative Distribution Function $F(x)$

For the distribution in the previous question, find the median $m$.

**a.** $m = \sqrt[3]{4} \approx 1.587$ ✓

**b.** $m = 1.0$

**c.** $m = \sqrt{2} \approx 1.414$

**d.** $m = 1.5$

Set $F(m) = 0.5 \Rightarrow \frac{m^3}{8} = 0.5 \Rightarrow m^3 = 4 \Rightarrow m = 4^{1/3} \approx 1.587$.

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**26** [Foundational] • Target: Cumulative Distribution Function $F(x)$

Given $F(x) = \frac{x^2 - 1}{8}$ for $1 \le x \le 3$, find the PDF $f(x)$ on this interval.

**a.** $f(x) = \frac{x}{4}$ ✓

**b.** $f(x) = \frac{x}{8}$

**c.** $f(x) = \frac{x^2}{8}$

**d.** $f(x) = \frac{2x - 1}{8}$

The PDF is the derivative of the CDF: $f(x) = F'(x) = \frac{d}{dx}\left[\frac{x^2 - 1}{8}\right] = \frac{2x}{8} = \frac{x}{4}$.

**27** [Foundational] • Target: Mean, Variance & Medians

Find the expected value $E(X)$ for the PDF $f(x) = 2x$ on $[0, 1]$.

**a.** $\frac{2}{3}$ ✓

**b.** $\frac{1}{2}$

**c.** $\frac{3}{4}$

**d.** $\frac{1}{3}$

$E(X) = \int_0^1 x f(x) dx = \int_0^1 2x^2 dx = [2x^3/3]_0^1 = 2/3$.

**28** [Foundational] • Target: Mean, Variance & Medians

For the distribution $f(x) = 2x$ on $[0, 1]$, calculate the variance $\mathrm{Var}(X)$.

**a.** $\frac{1}{18}$ ✓

**b.** $\frac{1}{9}$

**c.** $\frac{1}{12}$

**d.** $\frac{2}{9}$

$E(X^2) = \int_0^1 2x^3 dx = [x^4/2]_0^1 = 1/2$. $\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 1/2 - (2/3)^2 = 1/2 - 4/9 = 1/18$.

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Mean, Variance & Medians

Find the mode of the continuous distribution with PDF $f(x) = 12x^2(1 - x)$ for $0 \le x \le 1$.

**a.** $\frac{2}{3}$ ✓

**b.** $\frac{1}{2}$

**c.** $\frac{3}{4}$

**d.** $\frac{1}{3}$

The mode is where $f(x)$ is maximized: $f'(x) = 24x - 36x^2 = 12x(2 - 3x) = 0 \Rightarrow x = 2/3$.

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**30** [Intermediate] • Target: Piecewise Distributions & Examination Standards

A continuous random variable has PDF $f(x) = cx$ for $0 \le x < 2$ and $f(x) = c(4 - x)$ for $2 \le x \le 4$. Find $c$.

**a.** $\frac{1}{4}$ ✓

**b.** $\frac{1}{2}$

**c.** $\frac{1}{8}$

**d.** $1$

The PDF forms a symmetric triangle with base 4 and height $2c$. Area $= \frac{1}{2}(4)(2c) = 4c = 1 \Rightarrow c = 1/4$.

**31** [Intermediate] • Target: Piecewise Distributions & Examination Standards

For the symmetric triangular distribution in the previous question, what is $E(X)$?

**a.** **2** ✓

**b.** 1

**c.** 3

**d.** 2.5

By symmetry of the PDF about $x = 2$, the expected value must equal the line of symmetry: $E(X) = 2$.

**32** [Intermediate] • Target: Piecewise Distributions & Examination Standards

Find the lower quartile $Q_1$ for the triangular distribution in the previous question ($c = 1/4$).

**a.** $\sqrt{2} \approx 1.414$ ✓

**b.** $1.0$

**c.** $1.5$

**d.** $\sqrt{3} \approx 1.732$

For $x \le 2$: $F(x) = \int_0^x \frac{1}{4} t\,dt = \frac{x^2}{8}$. Set $F(Q_1) = 0.25 \Rightarrow \frac{Q_1^2}{8} = \frac{1}{4} \Rightarrow Q_1^2 = 2 \Rightarrow Q_1 = \sqrt{2} \approx 1.414$.

**33** [Intermediate] • Target: Piecewise Distributions & Examination Standards

The continuous random variable $X$ has probability density function given by $f(x) = \{k(4 - x^2), 0 \le x \le 2, ; 0, \text{otherwise}, \text{ where } k \text{ is a constant. Find the value of } k.$

**a.** $\frac{3}{16}$ ✓

**b.** $\frac{1}{4}$

**c.** $\frac{3}{8}$

**d.** $\frac{16}{3}$

For $f(x)$ to be a valid probability density function, the total area under the curve must equal 1: $\int_{-\infty}^{\infty} f(x)\,dx = 1$ Substitute the non-zero definition of $f(x)$: $\int_0^2 k(4 - x^2)\,dx = 1$ Factor out the constant $k$ and integrate term by term: $k\left[4x - \frac{x^3}{3}\right]_0^2 = 1$ Evaluate at the upper and lower limits: $k\left(\left(4(2) - \frac{2^3}{3}\right) - 0\right) = 1$ $k\left(8 - \frac{8}{3}\right) = 1$.

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**34** [Intermediate] • Target: Verifying a valid probability density function

A wire of length $4$ m is bent to form a rectangle with adjacent sides of lengths $X$ m and $(2 - X)$ m, where $X$ is a continuous random variable with probability density function $f(x) = \begin{cases} \frac{3}{8}x^2, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$ Given that the area of the rectangle is at least $\frac{3}{4}$ m$^2$, find the probability that the length of the shorter side is at most $\frac{2}{3}$ m.

**a.** $\frac{127}{351}$ ✓

**b.** $\frac{74}{351}$

**c.** $\frac{37}{351}$

**d.** $\frac{127}{640}$

Let $A$ denote the area of the rectangle and $L_{\min}$ denote the length of the shorter side. 1. **Determine the condition for $A \ge \frac{3}{4}$:** The area is given by $A = X(2 - X) = 2X - X^2$.
$2X - X^2 \ge \frac{3}{4} \Leftrightarrow X^2 - 2X + \frac{3}{4} \le 0 \Leftrightarrow (X - \frac{1}{2})(X - \frac{3}{2}) \le 0$ Thus, $A \ge \frac{3}{4}$ corresponds to the interval $E_1 = [\frac{1}{2}, \frac{3}{2}]$. Using the cumulative distribution function.

**35** [Intermediate] • Target: Finding an unknown constant in a pdf

A continuous random variable $X$ has probability density function given by $f(x) = \frac{3}{8}x^2$ for $0 \le x \le 2$, and $f(x) = 0$ otherwise. Find $\mathrm{P}(1 \le X \le 2)$.

**a.** $\frac{7}{8}$ ✓

**b.** $\frac{1}{8}$

**c.** $\frac{3}{4}$

**d.** $\frac{1}{2}$

To find the probability $\mathrm{P}(1 \le X \le 2)$, we integrate the probability density function $f(x)$ over the interval $[1, 2]$: $\mathrm{P}(1 \le X \le 2) = \int_1^2 \frac{3}{8}x^2 \, dx$ Find the antiderivative: $\int \frac{3}{8}x^2 \, dx = \frac{3}{8} \cdot \frac{x^3}{3} = \frac{x^3}{8}$ Evaluate between the limits $x = 1$ and $x = 2$: $\left[\frac{x^3}{8}\right]_1^2 = \frac{2^3}{8} - \frac{1^3}{8} = \frac{8}{8} - \frac{1}{8} = \frac{7}{8}$ Common errors include calculating t.

**36** [Intermediate] • Target: Finding probabilities by integrating the pdf

The continuous random variable $X$ has probability density function $f(x)$ defined by \[ f(x) = \begin{cases} kx & 0 \le x 1 \mid X < 2.5) \).

**a.** $\frac{9}{11}$ ✓

**b.** $\frac{10}{11}$

**c.** $\frac{3}{4}$

**d.** $\frac{9}{10}$

Step 1: Determine the constant $k$ using the total probability property $\int_{-\infty}^{\infty} f(x) \, dx = 1$.
$\int_0^2 kx \, dx + \int_2^3 k(6 - 2x) \, dx = 1$ Evaluating each integral: $\int_0^2 kx \, dx = k\left[\frac{x^2}{2}\right]_0^2 = 2k$
$\int_2^3 k(6 - 2x) \, dx = k[6x - x^2]_2^3 = k((18 - 9) - (12 - 4)) = k(9 - 8) = k$ Thus, $2k + k = 3k = 1 \Rightarrow k = \frac{1}{3}$. Step 2: Find $\mathrm{P}(X1 \mid X1)$ instead of.

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**37** [Intermediate] • Target: The cumulative distribution function

A continuous random variable X has PDF $f(x) = \frac{2}{10^2}x$ for $0 \le x \le 10$. Find the median of X in exact form.

**a.** $10\sqrt{2}$

**b.** $\frac{10}{2}$

**c.** $\frac{10\sqrt{2}}{2}$ ✓

**d.** $\frac{10\sqrt{2}}{4}$

$F(m) = (\frac{m}{10})^2 = \frac{1}{2} \Rightarrow m = \frac{10}{\sqrt{2}} = \frac{10\sqrt{2}}{2}$.

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Using the cdf to find probabilities • Cross-Lesson: 3.01 Advanced Trigonometry

A continuous random variable X has PDF $f(x) = \frac{2}{14^2}x$ for $0 \le x \le 14$. Find the median of X in exact form.

**a.** $14\sqrt{2}$

**b.** $\frac{14}{2}$

**c.** $\frac{14\sqrt{2}}{4}$

**d.** $\frac{14\sqrt{2}}{2}$ ✓

$F(m) = (\frac{m}{14})^2 = \frac{1}{2} \Rightarrow m = \frac{14}{\sqrt{2}} = \frac{14\sqrt{2}}{2}$.

**39** [Challenging] • Target: Expectation of a continuous random variable • Cross-Lesson: 4.01 Differentiation

A continuous random variable X has PDF $f(x) = \frac{2}{12^2}x$ for $0 \le x \le 12$. Find the median of X in exact form.

**a.** $\frac{12\sqrt{2}}{2}$ ✓

**b.** $\frac{12\sqrt{2}}{4}$

**c.** $\frac{12}{2}$

**d.** $12\sqrt{2}$

$F(m) = (\frac{m}{12})^2 = \frac{1}{2} \Rightarrow m = \frac{12}{\sqrt{2}} = \frac{12\sqrt{2}}{2}$.

**40** [Challenging] • Target: Variance of a continuous random variable • Cross-Lesson: 5.01 Integration

A continuous random variable X has PDF $f(x) = \frac{2}{2^2}x$ for $0 \le x \le 2$. Find the median of X in exact form.

**a.** $\frac{2\sqrt{2}}{2}$ ✓

**b.** $\frac{2\sqrt{2}}{4}$

**c.** $\frac{2}{2}$

**d.** $2\sqrt{2}$

$F(m) = (\frac{m}{2})^2 = \frac{1}{2} \Rightarrow m = \frac{2}{\sqrt{2}} = \frac{2\sqrt{2}}{2}$.

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**41** [Challenging] • Target: Median and quartiles • Cross-Lesson: 6.01 Sequences & Series

A continuous random variable X has PDF $f(x) = \frac{2}{6^2}x$ for $0 \le x \le 6$. Find the median of X in exact form.

**a.** $\frac{6\sqrt{2}}{4}$

**b.** $\frac{6}{2}$

**c.** $6\sqrt{2}$

**d.** $\frac{6\sqrt{2}}{2}$ ✓

$F(m) = (\frac{m}{6})^2 = \frac{1}{2} \Rightarrow m = \frac{6}{\sqrt{2}} = \frac{6\sqrt{2}}{2}$.

**42** [Challenging] • Target: A well-designed word problem: waiting time at a bus stop • Cross-Lesson: 7.01 Vectors

A continuous random variable X has PDF $f(x) = \frac{2}{8^2}x$ for $0 \le x \le 8$. Find the median of X in exact form.

**a.** $8\sqrt{2}$

**b.** $\frac{8\sqrt{2}}{4}$

**c.** $\frac{8}{2}$

**d.** $\frac{8\sqrt{2}}{2}$ ✓

$F(m) = (\frac{m}{8})^2 = \frac{1}{2} \Rightarrow m = \frac{8}{\sqrt{2}} = \frac{8\sqrt{2}}{2}$.

**43** [Challenging] • Target: Probability Density Functions (PDF) • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

A continuous random variable X has PDF $f(x) = \frac{2}{4^2}x$ for $0 \le x \le 4$. Find the median of X in exact form.

**a.** $\frac{4\sqrt{2}}{4}$

**b.** $\frac{4}{2}$

**c.** $\frac{4\sqrt{2}}{2}$ ✓

**d.** $4\sqrt{2}$

$F(m) = (\frac{m}{4})^2 = \frac{1}{2} \Rightarrow m = \frac{4}{\sqrt{2}} = \frac{4\sqrt{2}}{2}$.

**44** [Challenging] • Target: Cumulative Distribution Function $F(x)$ • Cross-Lesson: 9.01 Complex Numbers

A continuous random variable X has PDF $f(x) = \frac{2}{5^2}x^1$ for $0 \le x \le 5$. Find $\mathrm{Var}(X)$.

**a.** $\frac{10}{3}$

**b.** $\frac{25}{18}$ ✓

**c.** $\frac{25}{2}$

**d.** $\frac{43}{18}$

$E(X) = \frac{10}{3}$, $E(X^2) = \frac{25}{2}$. $\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \frac{25}{18}$.

**45** [Challenging] • Target: Mean, Variance & Medians • Cross-Lesson: 1.01 Functions: Modulus & Inverse

A continuous random variable X has PDF $f(x) = \frac{3}{6^3}x^2$ for $0 \le x \le 6$. Find $\mathrm{Var}(X)$.

**a.** $\frac{108}{5}$

**b.** $\frac{9}{2}$

**c.** $\frac{27}{20}$ ✓

**d.** $\frac{47}{20}$

$E(X) = \frac{9}{2}$, $E(X^2) = \frac{108}{5}$. $\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \frac{27}{20}$.

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